

Golden case 9: the reference is one of several recovered mappings

MAPPA search · 1 seed (default) + cut sweep · cap 2,000 · iso tolerance 1 · event threshold 0.5

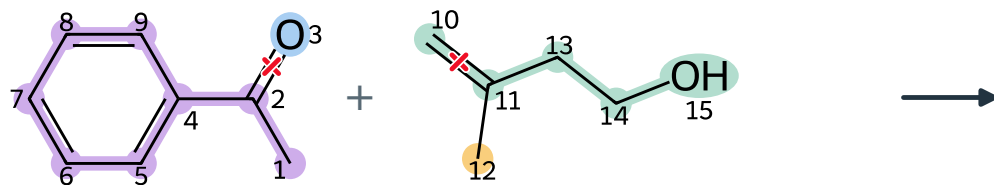
● ● ● Saved fragment groups

—~~×~~— Break / weaken

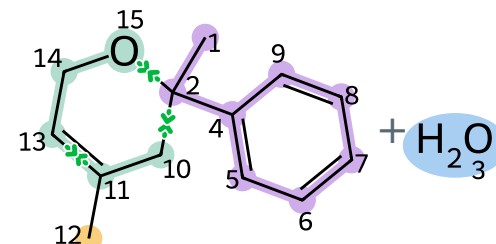
—~~×~~— Form / strengthen

A Ground truth (Golden reference) — recovered by MAPPA

5 heavy-atom changes



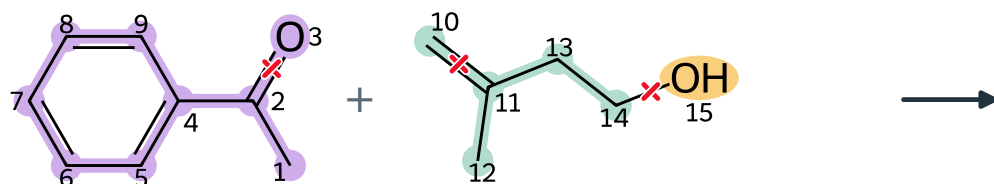
Reference: alcohol O₁₅ enters the ring; ketone O₃ leaves in water.



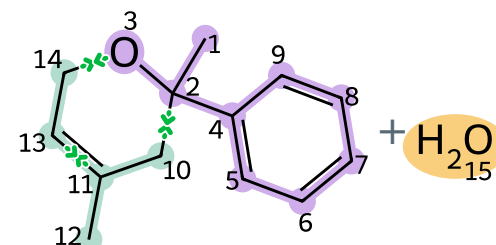
9 total events = 5 heavy + 4 involving H

B MAPPA alternative — different oxygen retained

6 heavy-atom changes



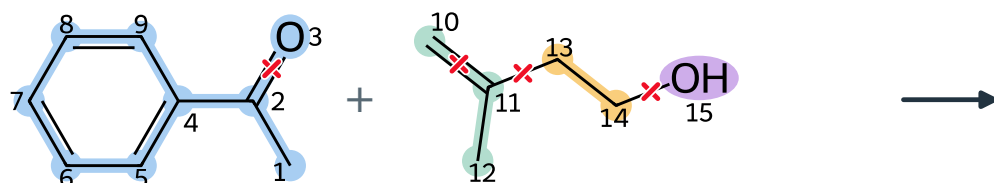
Alternative: ketone O₃ enters the ring; alcohol O₁₅ leaves in water.



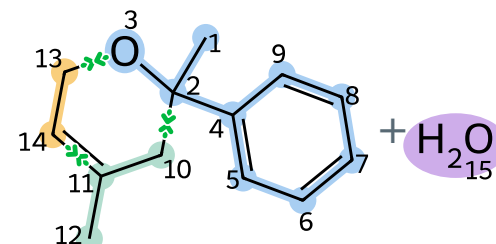
8 total events = 6 heavy + 2 involving H

C MAPPA alternative — oxygen and carbon correspondence change

7 heavy-atom changes



Alternative: O₃ enters the ring, with a different assignment of C₁₃ and C₁₄.



9 total events = 7 heavy + 2 involving H

Retaining alternatives matters: a lower total event count (B: 8) need not select the annotated mapping (A: 9).

Ground truth refers to Golden's heavy-atom annotation. H identities are unannotated; totals use each MAPPA witness.

Numbers are original Golden labels. H atoms are implicit. Selected witnesses do not imply exhaustive or minimum-event decoding.